Dr. Jagrati Talreja, PhD

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GitHub LinkedIn Portfolio

SUMMARY

AI Research Scientist and Machine Learning Engineer with 5+ years of experience in computer vision, deep learning, multimodal AI, and data analytics. Expertise in building CNN, Transformer, GAN, and diffusion-based systems for image enhancement, remote sensing, SAR-optical translation, and satellite/UAV data fusion. Strong background in Python, PyTorch, GPU training, experimental design, and research-to-production-style prototyping. Published in IEEE Transactions on Geoscience & Remote Sensing and experienced in federally funded NASA/NSF research environments.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS, Node.js, Bootstrap, MATLAB.
ML/AI: RAG, Agentic AI, CNNs, Vision Transformers, GANs, Diffusion Models, Autoencoders, PINNs, Transfer Learning, PyTorch, TensorFlow, Scikit-learn, OpenCV, Keras, XAI, Segmentation, Classification, Regression, Super-Resolution.
Data & Geospatial: Remote Sensing, Sentinel-1, Sentinel-2, UAV imagery, multimodal fusion, geospatial preprocessing, Google Earth Engine, Alaska Satellite Facility, ArcGIS Pro, SNAP.
Infra & Tools: CUDA, cuDNN, Jupyter, VS Code, Linux, WSL, Git, Overleaf, Office 365, PyCharm, Anaconda.
Hardware/Systems: NVIDIA A4000/A6000, A100/H100 clusters, FPGA, Raspberry Pi, SoC systems, Arduino.

EXPERIENCE

Postdoctoral Fellow / Co-Lead / Co-PI, NASA & NSF Funded Projects — Jan 2025 – Present
North Carolina A&T State University — *Greensboro, NC, USA*

- **Co-Principal Investigator** on **NASA** COMPASS proposal (2026) for AI-driven remote sensing.
- **Led** end-to-end development of multimodal deep learning pipelines for flood mapping across SAR and optical datasets, improving model performance and scalability on GPU clusters.
- **Designed and deployed** CNN/Transformer/GAN pipelines improving PSNR by +**2.2** dB (40% reduction in reconstruction error) on 20,000+ multimodal satellite image tiles using PyTorch on multi-GPU clusters.
- **Trained models** on A4000/A6000 GPU workstations for large-scale satellite data.
- Engineered scalable geospatial ML pipelines.
- **Collaborated** with NASA-affiliated teams and mentored students.
- **Developing** quantum-inspired ML pipelines, exploring spatial feature encoding for next-generation ML systems.

PhD Researcher, Electrical Engineering — Aug 2020 – 2024
Chulalongkorn University — *Bangkok, Thailand*

- **FPGA Super-Resolution:** Real-time deployment of a deep learning model for FPGA-based SoM **(Kria KV260)**.
- **Signature Verification:** Developed an offline authentication system using shape-based features to verify signatures.
- **Embedded Vision:** Developed an OpenCV-based contour detection and tracking system using Raspberry Pi.
- Mentored CV/DSP labs (Teaching Assistant).

Research Assistant / Intern — Jun 2018 – Jun 2019
Indian Institute of Technology — *Kanpur, India*

- **Disruption Detector**: Developed a MATLAB-based computer vision system that applies image registration and change detection to identify object displacement between pre- and post-event images.

Engineering Projects (Undergrad) — Aug 2015 – Dec 2018
Pranveer Singh Institute of Technology — *Uttar Pradesh, India*

- **Medi-Cart:** Developed an RFID-based healthcare system for patient identification, medication, and record tracking.
- **Robo Car:** Built an Arduino-based robotic vehicle controlled through real-time voice commands.
- **Aircraft Prototype:** Designed an aircraft prototype using a transmitter-receiver control system.
- **Android Apps:** Designed and developed interactive mobile applications using Java and Android Studio.
- **Software System:** Developed a Java- and MySQL-based platform for managing academic records and administrative data.

EDUCATION

PhD in Data Science & Artificial Intelligence (GPA 3.38) 2019 – 2024
Chulalongkorn University *Bangkok, Thailand*

Thesis: *Single Image Super-Resolution using Deep Attention Networks.*

- C2F Scholarship (PhD, 2019), Chulalongkorn University. ***(Highest PhD Scholarship in Thailand)***
- Overseas Research Experience Scholarship, 2024.
- 90th Anniversary Scholarship, Chulalongkorn University.

B.Tech in Electronics & Communication Engineering (GPA 3.24) 2015 – 2019
Pranveer Singh Institute of Technology *Kanpur, India*

- **Gold Medal** for excellence in Institutional Academia with **80.93%**.
- Final-year project resulted in an **Indian patent**.
- Selected among the **top 100 Engineering & Science projects** in Uttar Pradesh, India (2019).
- **Awarded** University and State Government Scholarships for 4 years.

ACHIEVEMENTS

- Graduate Record Examinations **(GRE)**: 304 (2020).
- International English Language Testing System Academic **(IELTS)**: 6.5 Overall (2019).
- Indian Institute of Technology Joint Entrance Examination **(IIT JEE)** Mains Qualified (2015).

CONFERENCES, SPEAKING & LEADERSHIP

- **Speaker**, International Geoscience and Remote Sensing Symposium (IGARSS), August 10-15, 2026.
- **Speaker**, American Society for Photogrammetry and Remote Sensing (ASPRS), February 15-19, 2026.
- **Speaker**, American Geophysical Union (AGU), December 15-19, 2025.
- **Speaker**, GIS Day, November 19, 2025.
- **Host**, IEEE R10 Robotics Competition for Managing Climate Change, Chulalongkorn University (2023).

PATENTS

- **SafePath Navigate to Safety**, US Application No. 63/929,945, 2025.
- **Medi-Cart: Telemetry using RFID**, Indian Patent No. 201811037266A, 2018.

PUBLICATIONS

- **J. Talreja**, T.S. Gebre, L.H. Beni, "FLoRA: Fusion-Latent for Optical Reconstruction and Flood Area Segmentation via Cross-Modal Multi-Task Distillation Network," IEEE Transactions on Geoscience & Remote Sensing, 2026.
- **J. Talreja**, T.S. Gebre, L.H. Beni, "DPolTransGAN: Scattering Guided Hybrid Transformer-GAN for SAR-Optical Image Translation using Dual Polarization SAR," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2026 .
- **J. Talreja**, T.S. Gebre, L.H. Beni, "Cross-Polarization Fusion of VV and VH SAR Observations for Improved Flood Mapping," *IGARSS*, 2026.
- **J. Talreja**, S. Aramvith, T. Onoye, "DANS: Deep Attention Network for Single-Image Super-Resolution," *IEEE Access*, 2023.
- **J. Talreja**, S. Aramvith, T. Onoye, "DHTCUN: Deep Hybrid Transformer Network for Single-Image Super-Resolution," *IEEE Access*, 2024.
- **J. Talreja**, S. Aramvith, T. Onoye, "XTNSR: Xception-Based Transformer Network for Single-Image Super-Resolution," *Complex & Intelligent Systems*, 2024.
- **J. Talreja**, L.H. Beni, "SARGANet: A GAN-U-Net Hybrid Framework for Super-Resolution of Sentinel-1 SAR Imagery," Earth Science Informatics, 2026.
- **J. Talreja**, L.H. Beni, "STCN-SR: A Spatial Transformer-CNN Framework for Enhancing Sentinel-2 Imagery," Geomatica, 2026.
- **J. Talreja**, T.S. Gebre, L.H. Beni, "Region8: Spatially-Aware Quantum Encoding for Flood Patch Classification in Multi-Modal Remote Sensing," Under Review.
- T.S. Gebre, **J. Talreja**, L.H. Beni, "Overcoming Physics Shock" in Earth Observation A Heteroscedastic Uncertainty Framework for PINN-based Flood Inference," Under Review.